
GAURAV MEHTA

Software Engineer II | C++ & Edge AI Specialist | US Citizen

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Specializing in On-Device AI optimizations and high-scale product integration. Delivered GenAI features for Copilot TV to 6M+ users and embedded inference pipelines at Qualcomm

PROFESSIONAL EXPERIENCE

MICROSOFT

July'24 – present

Software Engineer II, Microsoft AI

Copilot TV

- Engineered the **cloud-side** architecture for Copilot TV, orchestrating **capability negotiation and Bing search integration** to deliver context-aware **"What to Watch"** recommendations to 6M+ users
- Architected the hybrid execution environment** on Tizen, implementing a **public-IP proxy** and **local page deployment** to enable Copilot integration on constrained TV hardware
- Led technical alignment** across **10+ partner squads** (Bing, CMC, Product, Design) to define the roadmap, deliver the **CES 2024** showcase demo, and successfully scaling the launch to Samsung & LG devices globally

On-Device Model Execution

- Prototyped a "PhoneRAG" system** on Android, integrating **Phi-3 (3.8B)** and **CLIP embeddings** within a native C++ layer to validate retrieval-augmented generation on Snapdragon® edge hardware.
- Delivered the **inference pipeline** using **ONNX Runtime** and **Qualcomm QNN**, implementing **mixed-precision(16b/4b) quantization**, model splitting, and cached graph compilation to minimize initialization latency.
- Accelerated token generation throughput** from 0.8 to **2.7 TPS (NPU)** and **11 TPS (Hybrid)** by orchestrating heterogeneous execution, conducting deep performance benchmarking to inform **executive leadership** on power/thermal feasibility for future product roadmaps.

Tools – C++ (Native), Python, ONNX Runtime, Qualcomm QNN, Android, Tizen, C#, Bing Grounding API

QUALCOMM

Jul'21 – July'24

Software Engineer, ASE Team

Aware Secure Edge (ASE)

- Architected a secure IoT Marketplace** for deploying ML pipelines on Snapdragon edge devices, designing the **Feature Bundle Deployment** flow with **secure key provisioning** to decrypt models at runtime for multimedia use cases.
- Delivered the **License Service** (C++/gRPC), integrating with **ARM TrustZone (TEE)** to implement a **chunked decryption pipeline**, hardware-enforced licensing, and **secure telemetry**, ensuring tamper-proof execution of proprietary IP.
- Developed the **Feature Management Service** in **Golang**, utilizing the **Docker Go** package to implement resource-efficient container updates and distroless images. Built the **Device Registration** and **QAgent** tools to streamline fleet onboarding and ISV publishing workflows.
- Led the platform migration** from Linux Embedded to Android, re-architecting native services into **Stable AIDL interfaces** to ensure production compatibility with the Android Framework, presenting progress to senior leadership (SVP).

Tools – C++ (Native), Golang, Docker, Android (AIDL), ARM TrustZone, gRPC, REST, Redis, SQLite, Yocto, Linux, MQTT

Software Engineering Intern, IOT-ML Team**Deploying ML on Edge Devices**

- **Enabled hardware acceleration** on Snapdragon platforms by deploying **TensorFlow Lite** (v2.5) delegates to leverage **DSP/GPU** compute, significantly improving inference performance.
- **Pioneered the containerization** of TFLite applications using **Docker**, establishing the framework for scalable and standardized deployment on Qualcomm edge devices.

Tools - C++, Python, TFLite, Docker, CMake, DSP/GPU Shared Libraries, Python, dlib, OpenCV

EDUCATION & CREDENTIALS

M.S. Artificial Intelligence | University of Texas at Austin (Expected 2026)

- *Pursuing part-time alongside full-time employment*

B.E. Computer Science | BITS Pilani Hyderabad Campus (2021)

AWARDS

- **Qualcomm Distinguished Solution (2023)**: Invention accepted as a Trade Secret and published as a QDS
- **Finalist, Qualcomm Machine Learning Summit (2022)**: Selected to present "*Secure Deployment of IP to the Edge*" at the Qualcomm ML Summit '22
- **Runner Up, Qualcomm Tech Day (2022)**: Secured 2nd place company-wide for the presentation on "Deployment of Machine Learning to Edge Devices"
- **Qualcomm Rising Star Award (2022)**: Honored as a Rising Star among new joiners for significant contributions.